

Verification Roadmap

Your step-by-step guide to
becoming a skilled DV engineer.



The Beginning

When I started, it all felt overwhelming:

- UVM structure
- Complex protocols
- Debugging errors
- Understanding coverage metrics

I often wondered:

“Will I ever feel confident in the field?”

Good News: There's a clear path forward. Let's dive deep

Foundation

Step 1: Build your basics

- Master Systemverilog basics (classes, interfaces, randomization).
- Write a basic constrained random testbench.
- Practice using waveform debugging tools (Verdi, SimVision).

“Strong fundamentals will set you apart”

UVM&ENVIRONMENT

Step 2 : Learn UVM & Environment

- Understand UVM architecture : agents,drivers,monitors.
- Create a simple UVM testbench.
- Learn functional coverage and why it matters.
- Read and analyze seniors code.

“Start turning theory into practice.”

Protocol Mastery

Step 3 : Master Key protocols

- Choose 1-2 protocols : AXI,PCIe, AHB,etc.
- Write protocol-specific test scenarios.
- Understand scoreboards & checkers.

“ Protocols are the language of hardware.”

Advanced debug & Formal

Step 4 : Advanced Debugging & Formal verification

- Sharpen your debugging techniques across UVM layers.
- Learn Systemverilog Assertions (SVA) for catching corner case bugs.
- Get exposure to Formal Verification (Property checking & constraints)